NOTES 05: QUANTITATIVE VARIABLES

Stat 120 | Spring 2026

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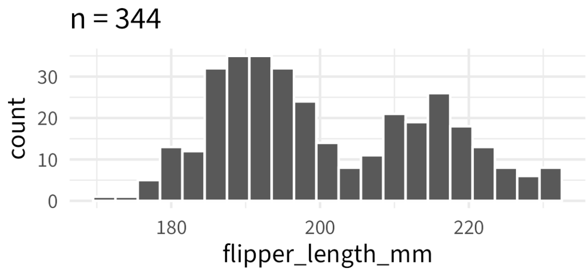
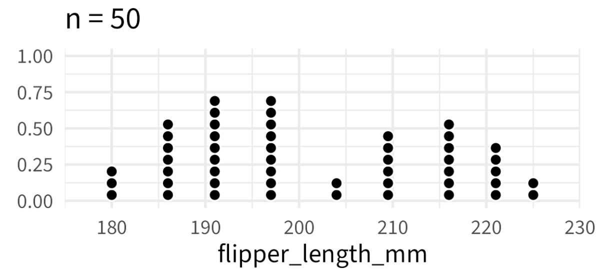
Quantitative variables are best visualized with a **histogram** (better for large samples) or **dotplot** (better for small samples)



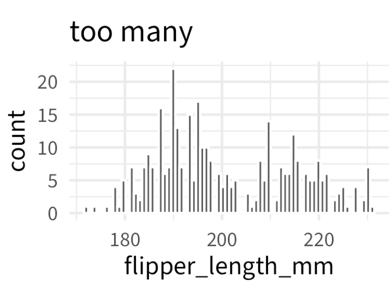


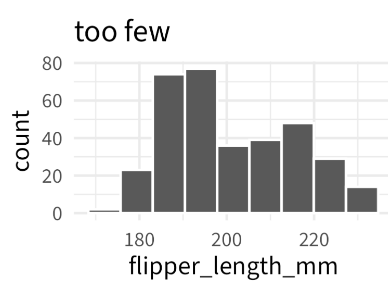
When describing quantitative variables, we typically care most about the **shape** and **center**.

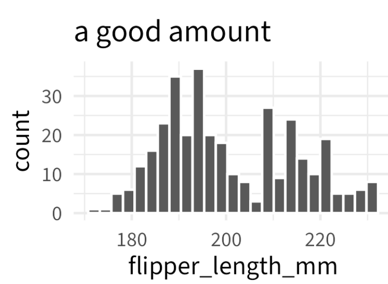
Skewed Right	Symmetric	Skewed Left
.		

It’s also important to note if the shape is **unimodal** or **multimodal**

When making a histogram, it’s important to choose an appropriate **number of bins**







There are various ways to describe the **center** of the distribution. The three most common are:

Mean: add up values and divide by *n*

- Sample:
- Population:
- $\bar{x} = \frac{x_1+x_2+\ldots+x_n}{n} = \frac{1}{n}\sum x_i$

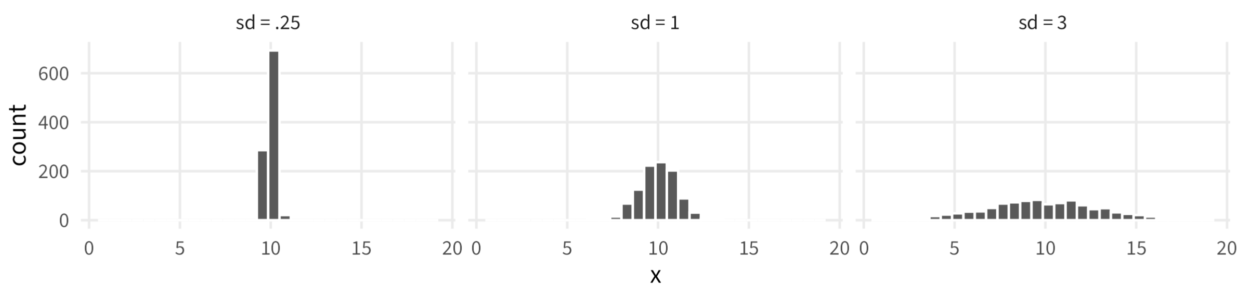
Median: midpoint

- odd *n*: middle number
- even *n*: mean of the middle 2 values

Mode: most common single value

- peak of the histogram
- can have more than 1!

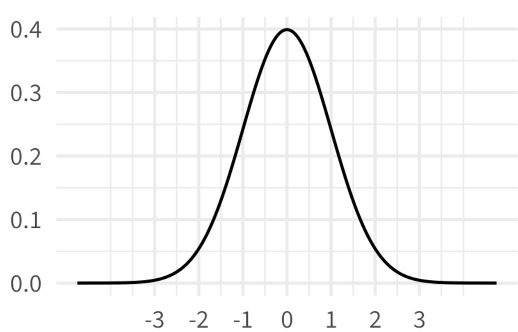
1 Standard Deviation: measure of spread



Standard Deviation: typical distance of a data point from the mean

$s = \sqrt{\frac{\sum(x_i-\bar{x})^2}{n-1}}$ R: `sd(...)`

The **68, 95, 99 rule** says that:

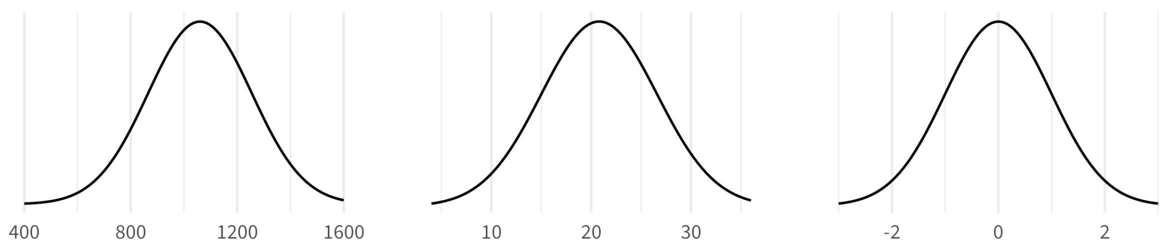


Example: Standardized test scores are often designed to have bell-curved distributions. The mean reported ACT score at Carleton is 33 with a standard deviation of 1.493. If we assume that Carleton's distribution is also symmetric and approximately bell-shaped, what interval gives the ACT score range of 95% of students?

2 Z-Scores

Z-Score: number of sd's away from the mean a data point is

Example: The nationwide average ACT score is 20.8 with a standard deviation of 5.8. The nationwide average SAT score is 1060 with a standard deviation of 195. A college admissions officer wants to determine which of two applicants scored better on their standardized test with respect to the other test-takers: Jordan, who earned an 1310 on their SAT, or Jim, who scored a 31 on his ACT?



3 Boxplots

A boxplot is a visualization of the **five number summary**, plus outliers.

Outlier: a data point that “stands out” from the rest

Boxplots are especially useful for comparing *quantitative variables* across different levels of a *categorical variable*.

4 Outliers and Skewed Data

Outliers can be annoying. We have three options for dealing with them:

1. Ignore them, and note that results may be impacted
2. Remove them, and note that results may be impacted (preferably include results from (1))
3. Transform the variable so they “matter” less